

COMPANION SLIDES

Chapter 2

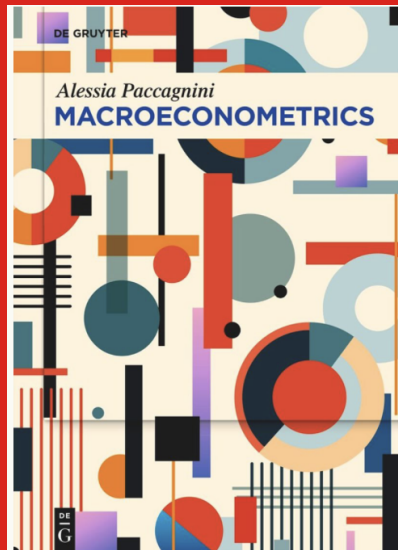
VAR Models

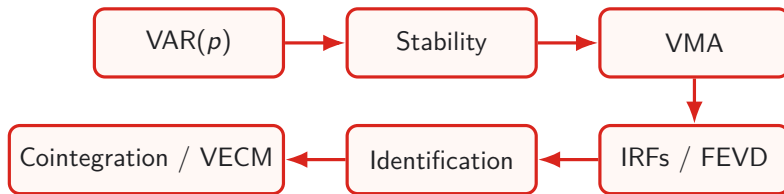
From Multivariate Dynamics to Impulse Responses

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Key takeaway

VARs turn a collection of interacting time series into a coherent dynamic system.

Macroeconomic systems are multivariate

- ▶ Variables interact contemporaneously and with lags.
- ▶ Shocks propagate through several channels over time.
- ▶ Common forces can move many variables together.

Questions a VAR can organize

What happens to output and inflation after an interest-rate innovation? How does an oil-price shock propagate? Which variables help predict others?

The VAR idea: treat every variable as potentially endogenous and let its dynamics depend on lags of the whole system.

- ▶ Write and interpret a $\text{VAR}(p)$.
- ▶ Check stability with companion-matrix eigenvalues.
- ▶ Derive the VMA representation.
- ▶ Estimate VARs by OLS and ML.
- ▶ Select lag length using information criteria.
- ▶ Test Granger causality.
- ▶ Recognize cointegration and use a VECM.
- ▶ Apply the Johansen procedure.
- ▶ Compute reduced-form IRFs and FEVDs.
- ▶ Understand why structural identification is required.

Standard representation

$$\mathbf{y}_t = \mathbf{c} + A_1 \mathbf{y}_{t-1} + \cdots + A_p \mathbf{y}_{t-p} + \boldsymbol{\varepsilon}_t, \quad E(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t') = \Sigma_{\varepsilon}.$$

- ▶ $\mathbf{y}_t$ is a $K \times 1$ vector of jointly modeled variables.
- ▶ Each coefficient matrix A_i captures own-lag and cross-variable effects.
- ▶ Reduced-form innovations may be contemporaneously correlated through Σ_{ε} .

Key takeaway

The simplicity of the VAR comes from using the same lagged information set in every equation.

Stability condition

A VAR(p) is stable when all eigenvalues of its companion matrix have modulus strictly smaller than one – equivalently, the roots of

$$\det(I_K - A_1z - A_2z^2 - \dots - A_pz^p) = 0$$

lie outside the unit circle.

If stable

- ▶ finite, time-invariant moments
- ▶ shocks dissipate
- ▶ a VMA representation exists

If not stable

- ▶ persistent stochastic trends may be present
- ▶ differencing or a VECM may be appropriate

For a stable VAR

$$\mathbf{y}_t = \boldsymbol{\mu} + \sum_{i=0}^{\infty} \boldsymbol{\Psi}_i \boldsymbol{\varepsilon}_{t-i}, \quad \boldsymbol{\Psi}_0 = \mathbf{I}_K.$$

- ▶ The matrices $\boldsymbol{\Psi}_i$ trace how reduced-form innovations propagate through the system.
- ▶ They are the building blocks for impulse responses and forecast error variance decompositions.
- ▶ The representation connects the VAR to the multivariate Wold decomposition.

VARMA(p, q)

$$A(L)\mathbf{y}_t = \mathbf{c} + B(L)\varepsilon_t.$$

Appeal

- ▶ combines AR and MA dynamics
- ▶ potentially more parsimonious

Why less common in applied macro

- ▶ identification is harder
- ▶ nonlinear estimation can be fragile
- ▶ interpretation is less transparent

Core question

Do lags of x_t improve forecasts of y_t once lags of y_t and the rest of the system are already included?

- ▶ Test joint restrictions on the relevant lag coefficients.
- ▶ Rejection means useful incremental predictive content.
- ▶ It does **not** by itself establish structural or causal identification.

OLS

Because every VAR equation contains the same regressors, each equation can be estimated separately by OLS.

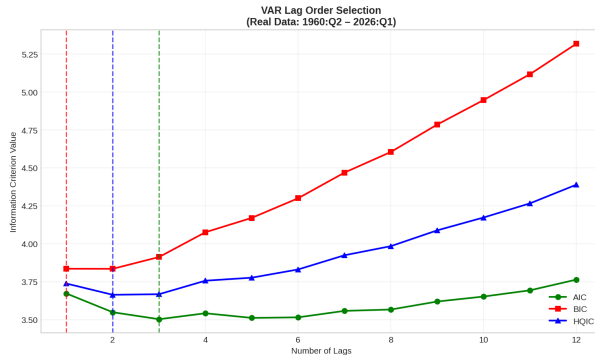
- ▶ simple and computationally convenient
- ▶ consistent under standard assumptions
- ▶ residual covariance is estimated after fitting the equations

Maximum likelihood

Under Gaussian innovations, ML provides an alternative system representation and connects naturally to likelihood-based inference.

Specification still matters: lag length, deterministic terms, sample, transformations and stability checks.

- ▶ AIC generally penalizes complexity less heavily.
- ▶ BIC favors more parsimonious specifications.
- ▶ HQ provides an intermediate penalty.
- ▶ Combine criteria with diagnostics and economic reasoning.



From VAR to VECM

For nonstationary variables with common stochastic trends, the VAR can be written as

$$\Delta \mathbf{y}_t = \Pi \mathbf{y}_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta \mathbf{y}_{t-i} + \varepsilon_t, \quad \Pi = \alpha \beta'.$$

- ▶ β contains the cointegrating relationships.
- ▶ α contains adjustment coefficients toward long-run equilibrium.
- ▶ The rank of Π determines the number of cointegrating vectors.

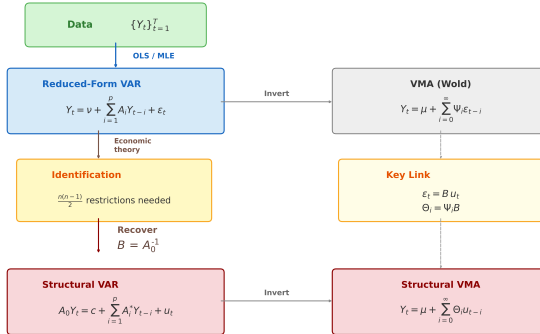
The Johansen procedure

- ▶ estimates the reduced-rank structure of Π ;
- ▶ uses trace and/or maximum-eigenvalue statistics;
- ▶ determines the cointegration rank r .

Representation choice

- ▶ stationary variables $\rightarrow$ VAR;
- ▶ nonstationary, no cointegration $\rightarrow$ differences;
- ▶ nonstationary + cointegration $\rightarrow$ VECM.

From Reduced-Form VAR to Structural VAR



Identification methods: Cholesky • Long-run restrictions • Sign restrictions • External instruments • Heteroskedasticity

Dynamic question

How does variable i respond over horizons $h = 0, 1, 2, \dots$ after an innovation in variable j ?

- ▶ In the VMA representation, Ψ_h contains the responses to reduced-form innovations.
- ▶ IRFs summarize magnitude, timing, persistence and sign of dynamic responses.
- ▶ For **causal** interpretation, the innovations must be mapped into structural shocks.

Forecast Error Variance Decomposition

Asks what fraction of the h -step forecast error variance of each variable is attributable to different shocks.

Historical Decomposition

Attributes observed historical movements to the accumulated contribution of identified shocks.

Key takeaway

IRFs describe propagation; FEVD describes relative importance; historical decomposition explains episodes.

Reduced-form innovations are not automatically structural shocks

$$\varepsilon_t = B\mathbf{u}_t, \quad E(\mathbf{u}_t\mathbf{u}_t') = I_K, \quad \Sigma_\varepsilon = BB'.$$

For K variables, Σ_ε supplies $K(K+1)/2$ unique moments but B has K^2 unknown elements. We therefore need $K(K-1)/2$ additional restrictions.

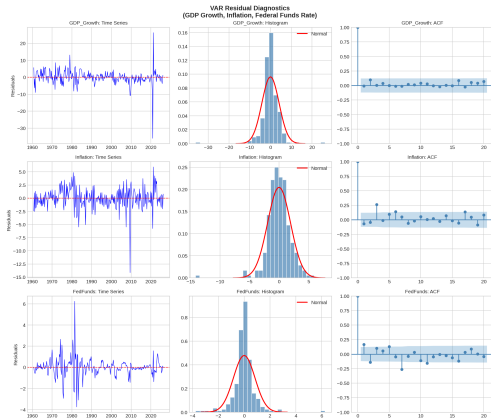
Key takeaway

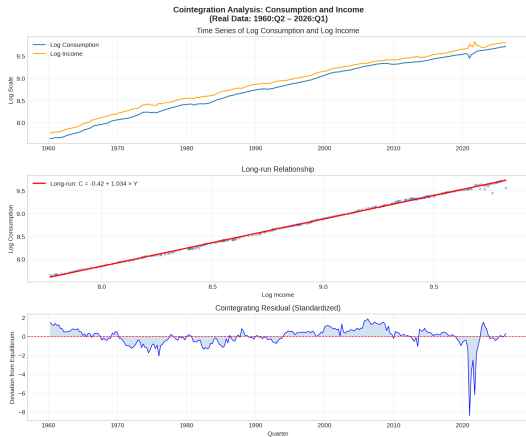
The data estimate the reduced form; economic or statistical restrictions recover interpretable structural shocks.

US quarterly data

- ▶ GDP growth
- ▶ inflation
- ▶ Federal Funds rate

The example walks through stationarity checks, lag selection, estimation, stability, residual diagnostics and Granger-causality analysis.



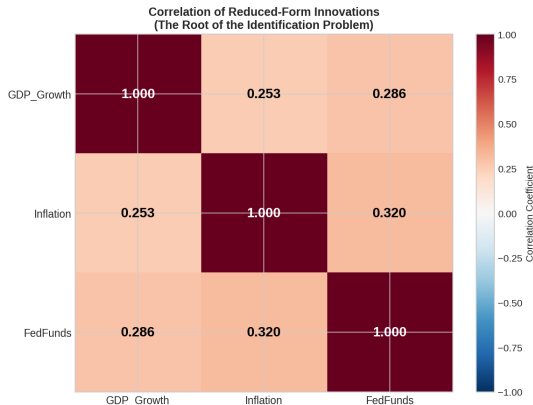


The multivariate VECM captures both the long-run equilibrium relation and potentially two-sided short-run adjustment.

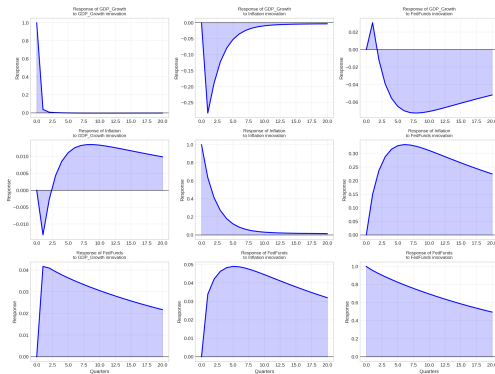
Estimated residual correlations

The book example reports positive contemporaneous correlations across the reduced-form innovations, with inflation–Fed Funds the strongest at about 0.32.

This makes a residual innovation an ambiguous mixture of structural forces.



Reduced-Form Impulse Response Functions
 □ NOT Economically Identified — These Are Statistical Responses Only



WARNING: These are REDUCED-FORM IRFs to correlated innovations. They confound multiple structural effects and cannot be given causal interpretation.

These responses describe propagation of reduced-form residual innovations. Because those innovations are correlated, structural interpretation requires additional identifying restrictions.



- ▶ Check whether the representation is appropriate before interpreting dynamics.
- ▶ Report uncertainty for IRFs and variance decompositions.
- ▶ Be explicit about identifying assumptions when moving from reduced-form to structural analysis.

1. VARs model **joint dynamics and feedback** without requiring a fully specified structural model at the estimation stage.
2. Stability gives the VMA representation that underpins dynamic analysis.
3. Cointegrated nonstationary systems call for the **VECM** representation.
4. Reduced-form IRFs are informative but are not automatically causal.
5. Structural interpretation requires **identification** – the bridge to Chapter 4.

Chapter 2 complete

VAR Models

Companion slides for
Macroeconometrics

Alessia Paccagnini | De Gruyter, 2026

Next: Local Projections